Ziru Chen

Curriculum Vitae

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Interests

I am broadly interested in *AI agents*, *language models*, and *machine learning*. My recent research focuses on (1) **test-time computation** of language models, especially **planning, self-verification, and post-training** and (2) **self-improving coding agents** for long-horizon, real-world tasks, such as data management & analysis, professional work, and AI-driven scientific research.

Education

The Ohio State University, Columbus, OH — Aug 2022 – May 2027
Ph.D. in Computer Science and Engineering — **GPA: 3.92 / 4.0**
Thesis: *Toward Self-Improving Language Models of Code with Test-Time Planning*

The Ohio State University, Columbus, OH — Aug 2026
M.S. in Computer Science and Engineering
Awarded for Ph.D. Candidacy

The Ohio State University, Columbus, OH — Aug 2018 – May 2022
B.S. in Computer and Information Science, Math Minor — **GPA: 3.97 / 4.0**
Summa Cum Laude

Experiences

Research Assistant, The Ohio State University — Dec 2020 – Present
Advisor: Dr. Huan Sun
- LLM Planning with Test-Time Computation (**ACL**'24)
- Language Agents for Coding and Scientific Data Analysis (**ICLR**'25, **EMNLP**'25)
- Interactive Text-to-SQL Generation (**ACL**'23, **EMNLP**'23, **EMNLP Findings**'23)
- User-Centered Task-Oriented Dialogue System (**Alexa Prize**'22, **3rd Place** Honor)

Research Intern, Google Cloud AI — May 2026 – Aug 2026
Mentors: Mr. Honglong Cai, Ms. Ramya Sree Boppana
- Self-Evolving Coding Agent for Long-Horizon Professional Work (In Submission)

Research Intern, Microsoft AI — May 2025 – Aug 2025
Mentors: Dr. Yujia Xie, Dr. Dongdong Chen
- Reinforcement Learning for Multi-Turn Code Generation (**TMLR**'26)

Honors

University Fellowship, Graduate School, OSU — 2022–2023, 2026–2027
3rd Place (Student Co-lead), 1st Alexa Prize Taskbot Challenge, Amazon — 2022
(10 participant teams selected worldwide out of 125 applications; The only US team in top-3)
Undergraduate Research Award, CSE Department, OSU — 2022
CIS Undergrad Scholarship, CSE Department, OSU — 2021

Publications

[19] *COBALT: Contextual Bandit Learning with Offline Trajectories for Multi-Turn Code Generation*
Ziru Chen, Dongdong Chen, Ruinan Jin, Yingbin Liang, Yujia Xie*, Huan Sun* (* equal advising)
Transactions on Machine Learning Research (TMLR 2026, Journal)
The 5th Deep Learning for Code Workshop at ICML 2026 (DL4C 2026, Long)

[18] *D3-Gym: Constructing Real-World Verifiable Environments for Data-Driven Discovery*
Hanane Nour Moussa, Yifei Li, Zhuoyang Li, Yankai Yang, Cheng Tang, Tianshu Zhang, Nesreen K. Ahmed, Ali Payani, **Ziru Chen**, Huan Sun
The 2026 Conference on Empirical Methods in Natural Language Processing (EMNLP 2026, Long)

[17] *Holistic Agent Leaderboard: The Missing Infrastructure for AI Agent Evaluation*
Sayash Kapoor, Benedikt Stroebl, Peter Kirgis, Nitya Nadgir, Zachary S Siegel, Boyi Wei, Tianci Xue, **Ziru Chen**, Felix Chen, Saiteja Utpala, Franck Ndzomga, Dheeraj Oruganty, Sophie Luskin, Kangheng Liu, Botao Yu, Amit Arora, Dongyoon Hahm, Harsh Trivedi, Huan Sun, Juyong Lee, Tengjun Jin, Yifan Mai, Yifei Zhou, Yuxuan Zhu, Rishi Bommasani, Daniel Kang, Dawn Song, Peter Henderson, Yu Su, Percy Liang, Arvind Narayanan

The Fourteenth International Conference on Learning Representations (ICLR 2026, Long)

[16] *Agent Data Protocol: Unifying Datasets for Diverse, Effective Fine-tuning of LLM Agents*
Yueqi Song, Ketan Ramaneti, Zaid Sheikh, **Ziru Chen**, Boyu Gou, Tianbao Xie, Yiheng Xu, Danyang Zhang, Apurva Gandhi, Fan Yang, Joseph Liu, Tianyue Ou, Zhihao Yuan, Frank F. Xu, Shuyan Zhou, Xingyao Wang, Xiang Yue, Tao Yu, Huan Sun, Yu Su, Graham Neubig
The Fourteenth International Conference on Learning Representations (ICLR 2026, Long)
Oral Presentation (Top 1% out of 19.8K Submissions)

[15] *Large Language Models Achieve Gold Medal Performance at International Astronomy & Astrophysics Olympiad*
Lucas Carrit Delgado Pinheiro*, **Ziru Chen***, Bruno Caixeta Piazza, Ness Shroff, Yingbin Liang, Yuan-Sen Ting, Huan Sun (* equal contribution)
Nature Astronomy (Nat Astron 2026, Journal)

[14] *AutoSDT: Scaling Data-Driven Discovery Tasks Toward Open Co-Scientists*
Yifei Li*, Hanane Nour Moussa*, **Ziru Chen**, Shijie Chen, Botao Yu, Mingyi Xue, Benjamin Burns, Tzu-Yao Chiu, Vishal Dey, Zitong Lu, Chen Wei, Qianheng Zhang, Tianyu Zhang, Song Gao, Xuhui Huang, Xia Ning, Nesreen K. Ahmed, Ali Payani, Huan Sun (* equal contribution)
The 2025 Conference on Empirical Methods in Natural Language Processing (EMNLP 2025, Long)
Best Paper Award, The first LLM for Scientific Discovery Workshop at COLM 2025

[13] *GeoAnalystBench: A GeoAI benchmark for assessing large language models for spatial analysis workflow and code generation*
Qianheng Zhang, Song Gao, Chen Wei, Yibo Zhao, Ying Nie, **Ziru Chen**, Shijie Chen, Yu Su, Huan Sun
Transactions in GIS 2025 (TGIS 2025, Journal)

[12] *ScienceAgentBench: Toward Rigorous Assessment of Language Agents for Data-Driven Scientific Discovery*
Ziru Chen, Shijie Chen, Yuting Ning, Qianheng Zhang, Boshi Wang, Botao Yu, Yifei Li, Zeyi Liao, Chen Wei, Zitong Lu, Vishal Dey, Mingyi Xue, Frazier N. Baker, Benjamin Burns, Daniel Adu-Ampratwum, Xuhui Huang, Xia Ning, Song Gao, Yu Su, Huan Sun
The Thirteenth International Conference on Learning Representations (ICLR 2025, Long)
Covered by Nature News. 25.6K+ downloads on Huggingface by October 9, 2026.

[11] *Tooling or Not Tooling? The Impact of Tools on Language Agents for Chemistry Problem Solving*
Botao Yu, Frazier N. Baker, **Ziru Chen**, Garrett Herb, Boyu Gou, Daniel Adu-Ampratwum, Xia Ning, Huan Sun
Findings of the Association for Computational Linguistics: NAACL 2025 (Findings 2025, Short)

[10] *When is Tree Search Useful for LLM Planning? It Depends on the Discriminator*
Ziru Chen, Michael White, Raymond Mooney, Ali Payani, Yu Su, Huan Sun
The 62nd Annual Meeting of the Association for Computational Linguistics (ACL 2024, Long)

[9] *eCeLLM: Generalizing Large Language Models for E-commerce from Large-scale, High-quality Instruction Data*
Bo Peng*, Xinyi Ling*, **Ziru Chen**, Huan Sun, Xia Ning (* equal contribution)
Proceedings of the 41st International Conference on Machine Learning (ICML 2024, Long)
Our ECInstruct dataset is recommended at Amazon KDD Cup 2024.

[8] *Automatic Evaluation of Attribution by Large Language Models*
Xiang Yue, Boshi Wang, **Ziru Chen**, Kai Zhang, Yu Su, Huan Sun
Findings of the Association for Computational Linguistics: EMNLP 2023 (Findings 2023, Long)

[7] *Error Detection for Text-to-SQL Semantic Parsing*
Shijie Chen, **Ziru Chen**, Huan Sun, Yu Su
Findings of the Association for Computational Linguistics: EMNLP 2023 (Findings 2023, Long)

[6] *Exploring Chain-of-Thought Style Prompting for Text-to-SQL*

Chang-Yu Tai*, **Ziru Chen***, Tianshu Zhang, Xiang Deng, Huan Sun (* equal contribution)
The 2023 Conference on Empirical Methods in Natural Language Processing (EMNLP 2023, Long)

[5] *Roll Up Your Sleeves: Working with a Collaborative and Engaging Task-Oriented Dialogue System*
Lingbo Mo, Shijie Chen, **Ziru Chen**, Xiang Deng, Ashley Lewis, Sunit Singh, Samuel Stevens, Chang-Yu Tai, Zhen Wang, Xiang Yue, Tianshu Zhang, Yu Su, Huan Sun
The 24th Meeting of the Special Interest Group on Discourse and Dialogue (SIGDIAL 2023, Demo)

[4] *Text-to-SQL Error Correction with Language Models of Code*
Ziru Chen, Shijie Chen, Michael White, Raymond Mooney, Ali Payani, Jayanth Srinivasa, Yu Su, Huan Sun
The 61st Annual Meeting of the Association for Computational Linguistics (ACL 2023, Short)

[3] *SalsaBot: Towards a Robust and Generalizable Embodied Agent*
Chan Hee Song*, Jiaman Wu*, Ju-Seung Byun, Zexin Xu, Vardaan Pahuja, Goonmeet Bajaj, Samuel Stevens, **Ziru Chen**, Yu Su (* equal contribution)
The 1st Proceedings of Alexa Prize Simbot Challenge (Alexa Prize 2023, Long)
The Embodied AI 2023 Workshop (EmbodiedAI 2023, Short)

[2] *Error Detection for Interactive Text-to-SQL Semantic Parsing*
Shijie Chen, **Ziru Chen**, Huan Sun, Yu Su
The 2nd Workshop on Interactive Learning for Natural Language Processing (InterNLP 2022, Short)

[1] *Bootstrapping a User-Centered Task-Oriented Dialogue System*
Shijie Chen*, **Ziru Chen***, Xiang Deng, Ashley Lewis, Lingbo Mo, Samuel Stevens, Zhen Wang, Xiang Yue, Tianshu Zhang, Yu Su, Huan Sun (* equal contribution)
The 1st Proceedings of Alexa Prize Taskbot Challenge (Alexa Prize 2022, Long)
Our TacoBot is the 3rd winner and deployed in Amazon Alexa for one year.

Teaching

CSE 5525 Speech and Language Processing, Undergraduate & Graduate Level
Teaching Assistant, The Ohio State University (Autumn 2021; Spring 2022)

CSE 2321 Foundations I: Discrete Structures, Undergraduate Level
Teaching Assistant, The Ohio State University (Autumn 2019; Spring, Autumn 2020)

Service

Recognitions: EMNLP 2024 (Outstanding Reviewer), ICML 2025 (Sliver Reviewer)
Program Committee: AAAI (2023, 2024, 2025), ACL Rolling Review (2024, 2025, 2026), COLM (2025, 2026), ICLR (2025, 2026, 2027), ICML (2026), NeurIPS (2025)
Journal Reviewer: ACM Computing Surveys (2026)
Workshop Reviewer: Reasoning and Planning for LLMs (ICLR 2025), LLM for Scientific Discovery (COLM 2025, COLM 2026), Deep Learning for Code (ICML 2026)

Skills

ML/LLM Programming
Pytorch, Huggingface, vLLM, OpenRLHF, veRL, asyncio, Ray
Languages
*Chinese (Native), English (Fluent), Japanese (Proficient; **JLPT N2**), Korean (Elementary)*